CS5365: Principles of Multi-processor Systems

Summer 2026

(Subject to change)

Project Requirements:

- Projects are individual.
- Each project must satisfy all of the following mandatory requirements:
  1. *Match with the course topics (examples are listed below).*
  2. *Perform a simulation (using any programming software or software simulators).*
- **Project presentation** must include a **video (~10 minutes duration)** in mp4 format to be provided to the instructor by **June 26th**. If the video file is large (>20 MB), instead of emailing the video, use a file sharing system such as google drive, with public sharing access of the video file and email the link to download the video file. All project videos will be uploaded to the instructor's YouTube Channel for this course: https://www.youtube.com/@CyberPhysicalSystemsTTU
- **Project report** should follow the sections of a typical conference paper. It can be formatted as a conference paper or a report. Project report must be submitted via RaiderCanvas by **July 2nd**.

Example topics of projects:

1. *Pipeline processor simulation*
2. *Memory architecture simulation*
3. *Vector processor/GPU simulation*
4. *Superscalar processor simulation*
5. *Multi-processor system simulation/Parallel computing*
6. *ISA RISC-V simulator (e.g., pre-fetching, Assembly ⇔ Machine code, TLM)*
7. *Ternary processor / computing simulation*
8. *Neuromorphic processor simulation*

Example software simulators:

1. *SiFive*
2. *SPIM*
3. *CocoTB*
4. *Spike*
5. *Gem5*
6. *QEMU*
7. *Sim-V*
8. *SimPy*
9. *Paessler*
10. *uCsim*